

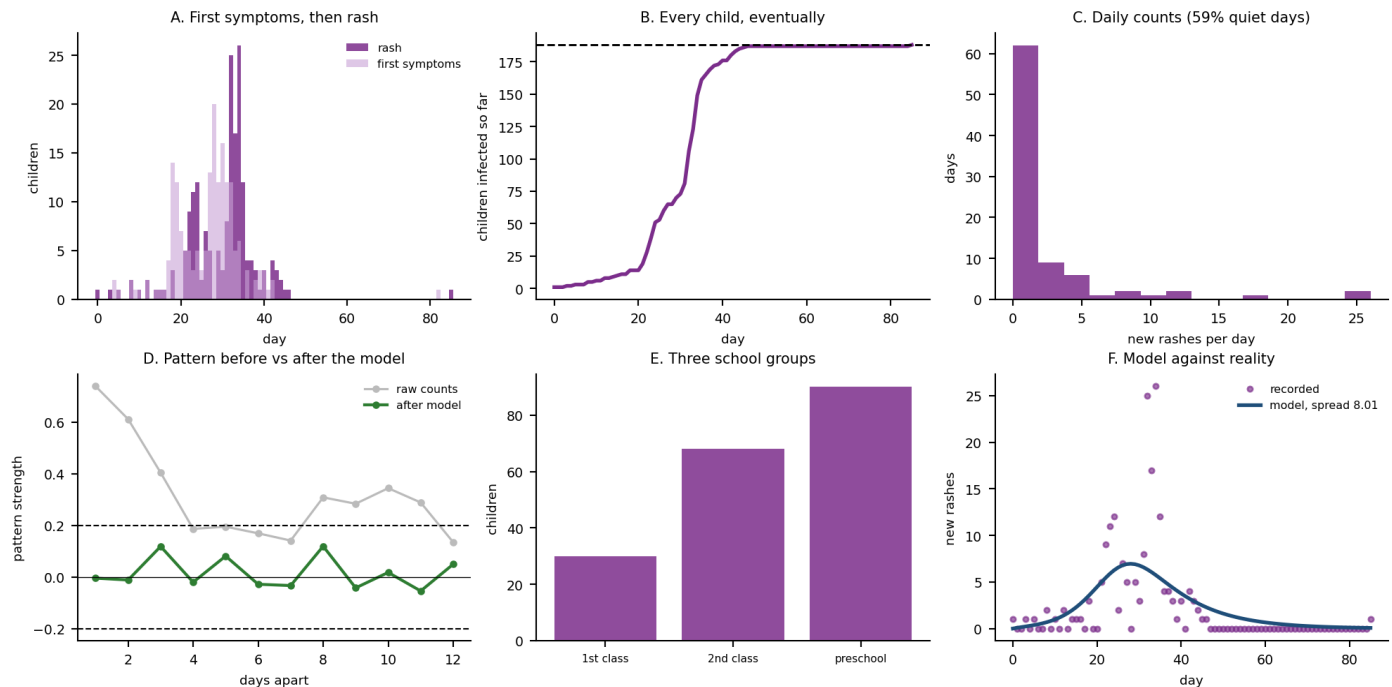
Measles — Hagelloch, Germany, 1861

Dossier 2 of 4 · every child in one village · Pfeilsticker (1863); Neal & Roberts (2004) Biostatistics 5:249

Findings, decisions, and where each one stands

Finding	Value in RAW data	Concern	Decision taken	Status in CANONICAL
Two symptom dates per child	first symptoms and rash, ~4 days apart	which one is the outbreak record?	use rash date — recorded for all 188 children	RESOLVED — 188 rashes = 188 children
Counts jump around	variance/mean 10.64	plain count model would be too rigid	use the forgiving (negative binomial) count	RESOLVED — model matched to it
Many quiet days	59% of days with no new rash	could look like missing records	none needed — village watched continuously, zeros are real	RESOLVED — all 86 days kept
Days follow one another	lag-1 pattern +0.739	might need a change-over-time allowance	compare to NEW rashes per day, same fix as Ebola	RESOLVED — -0.005 after the model
Everyone caught it	188 of 188 (100%)	the usual 'how many caught it' shortcut breaks at 100%	estimate contagiousness from the timing instead	HANDLED — shortcut skipped for this outbreak
Curve is spikier than the model	peak 26 vs model ~7	village mixes by household and classroom, not evenly	absorbed by the forgiving count; documented as a known limit	DISCLOSED — see caption on panel F

This shows that four concerns fully resolved, one handled a different way, one honestly disclosed. The disclosed one does not affect the timing, only how sharp the peak looks.



This shows that panel B is the striking one: the running total climbs to 188 and flattens, meaning literally every child in the village caught measles. That is what a disease with no vaccine and no prior immunity does to a closed community.

What the model will estimate

Quantity	Pre-fit estimate	Published range	Reading
Spread score (R0)	8.01	12 – 18 classically; 1 – 770 across settings (Guerra 2017)	below the classic band, as expected when a model assumes even mixing in a village with schools
Share who died	6.4% (12 of 188)	counted directly from the records	high by modern standards; no vaccine and no antibiotics in 1861
Usable observations	86 days	—	middle of the four outbreaks

This shows that the estimate sits a little under the textbook figure. That is a known consequence of the even-mixing assumption, not a sign the data or the model is wrong — and it is the interval, not this single number, that should be compared.

VERDICT — CLEARED. Timing is captured cleanly and nothing is left unexplained. The one imperfection, a smoother peak than reality, is understood, harmless to the timing, and stated openly. One of the three carrying the headline comparison.